PAULINA LILIAN RUÍZ SERVÍN

Electrical & Mechatronics Engineering Student · Embedded Systems, Electronics, Control & Automation

Personal Details



Location	Zittau, Saxony, Germany
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Nationality	Mexican
Availability	from March 2027, full-time, up to 6 months

Profile

Electrical and Mechatronics Engineering student (DHIK double degree) with hands-on experience in embedded systems, electronics, control and automation: C/C++ firmware for ESP32 with CAN, SPI, I2C and UART, PCB design in Altium, modelling in MATLAB/Simulink, PLC programming in TIA Portal with digital twin and virtual commissioning, and laboratory testing and verification; test automation in Python. Former captain of a 30-member racing team.

Education

B.Eng. Electrical Engineering (DHIK double degree) · Hochschule Zittau/Görlitz **09.2026 – 12.2027**
Zittau, Germany · Current courses: Microcontrollers, Image Processing, Advanced Control Theory

B.Sc. Mechatronics Engineering (fully taught in English) · Tecnológico de Monterrey (ITESM) **08.2023 – 12.2027**
Mexico City · GPA 92/100 (≈ 1.8 on the German scale) · Merit scholarship (30 %)

Engineering Experience

Team Captain, Electronics Lead & Mentor · EcoVolt CCM EV Racing Team **08.2023 – 08.2026**
Shell Eco-marathon · Mexico City

- Designed the team's telemetry PCBs in Altium Designer (schematic, layout, BOM): the Brazil 2025 board, main contributor to US 2026 and co-designer of Brazil 2026; brought them up and debugged them on the vehicle.
- Linked the in-house motor controller to the telemetry system via CAN (incl. transceiver circuit) and wrote C++ ESP32 firmware reading gyroscope, accelerometer, GPS, current/voltage sensors and ADCs (SPI, I2C, UART), logging to SD.
- Achieved bidirectional motor-current measurement, verified with oscilloscope and multimeter, by developing an INA240 shunt current sensor from a self-etched prototype to a manufactured board.
- Built a model-based understanding of the BLDC drive by performing measurements and contributing to the transfer functions of its current and speed loops in MATLAB/Simulink.
- Selected the components of a supercapacitor energy reserve (≈ 80 s backup) by characterizing transistors and Zener, Schottky and rectifier diodes in the lab; mentored the teammate who built it.
- Proved with measured data that constant throttle beats pulse-and-glide by building, with the team, a Python HIL test bench with a real traction motor and a driving simulator.
- Ran the driver ergonomics analysis in Process Simulate (virtual human, RULA), contributing to the Vehicle Design Award; built a QR-code tool inventory pilot in Python/OpenCV.
- Led the ∼30-member team as captain (2025–2026), developing an energy-efficient EV that achieved ∼200 km/kWh at ∼28 km/h. **Awards:** 1st (2025) and 2nd place (2026) Data & Telemetry Award; Vehicle Design Award (2026); Altair Honorable Mention (2025).

Visiting Student – Microfabrication · Massachusetts Institute of Technology (MIT.nano Lab) **10.2025**
Cambridge, USA

- Fabricated a solar cell, a MEMS cantilever and a microfluidic mixer in the cleanroom (photolithography, PECVD/PVD, etching) and characterized them: IV curve of the solar cell, mechanical force–displacement test of the cantilever and flow tests of the mixer.

Publication

Real-Time Telemetry and Computational Intelligence Pipeline for Energy-Efficient Driving of an Electric Eco-Marathon Vehicle **09.2026**

IEEE SAIBOT 2026 (Latin American School on Computational Intelligence and Robotics) · accepted

P. L. Ruíz-Servín, J.D. González-Fernández, A. Flores-Mendoza, J. Izquierdo-Reyes

Projects

Digital Twin & Virtual Commissioning of a CNC Cell · ORZA Tech / Tec de Monterrey **02.2026 – 06.2026**

Team project, 3 people; own station · Mexico City

- Tested the cell before any hardware existed by programming a 35-step PLC sequence in TIA Portal (virtual S7-1500, interlocks, manual/automatic modes, WinCC HMI), connecting a C# sensor via Modbus TCP and coupling everything with a Process Simulate digital twin and a virtual UR robot controller.
- Implemented recipes A–D (FIFO), OK/rework/scrap quality routing and KPI tracking (yield, scrap, micro-OEE) with a 100-cycle log; built the global SCADA to monitor the cell.
- Connected the UR5 through a virtual robot controller with a 7-signal handshake, wrote the CAM program of my station (6 tools, cycle time $\approx$ 12.8 min) and set up optical inspection with Cognex In-Sight (validated with a caliper, $\pm$3 %).

SCARA Robot (4 DoF): Control PCB, Motion Control & Vision · Tec de Monterrey **02.2026 – 06.2026**

Team project, 4 people · Mexico City

- Selected motors and drivers, designed the control PCB in Altium (ESP32, drivers, encoders), modelled joint PID control in MATLAB/Simulink and validated the homing sensors (limit switches, AS5600).
- Built the Python vision pipeline: ArUco/homography calibration to mm, YOLOv11n (mAP@0.5 = 0.72) and MQTT to the ESP32. Result: 5 of 6 parts sorted autonomously.

Measurement Electronics & Firmware: Viscosity System · Endress+Hauser / Tec de Monterrey **08.2025 – 12.2025**

Team project, 4 people · Mexico City

- Reliably separated fluids of 252 cP and 1676 cP through the DC motor current by co-developing a two-stage op-amp chain (1 Ω shunt, gain $\approx$ 400), simulated in Multisim and verified on the board.
- Co-programmed the system's conveyors with OpenPLC (ladder logic, Arduino Mega) and IR sensors.
- Designed all three PCBs in Altium (ESP32 master/slave to IPC-2221, amplifier board); co-wrote the C++ firmware: digital filters, PID control of the DC motors, I2C, UART, Bluetooth to a LabVIEW HMI.

Scissor Lift for an Automated Guided Vehicle (AGV) · Tec de Monterrey **02.2025 – 06.2025**

Team project, 5 people · Mexico City

- Built a load-cell amplifier without a dedicated IC (Wheatstone bridge, op-amp stages, RC low-pass) mapping 0–5 kg linearly to 0–3 V for the ESP32 ADC; designed the mechanism in SolidWorks with static analysis (max. deflection 0.28 mm); frame load-tested above 100 kg.

Skills

Embedded Systems	ESP32, ESP32-C3, Arduino, ESP-IDF, FreeRTOS, PlatformIO, C/C++
Communication	CAN (incl. transceiver), I2C, SPI, UART, Bluetooth, Wi-Fi/MQTT, Modbus TCP
Electronics & PCB	Op-amp signal conditioning, shunt current sensing, component characterization, digital logic (coursework); Altium Designer, IPC-2221, BOMs; SMD/THT soldering, PCB etching
Control & Simulation	PID, BLDC/DC/stepper motors, encoders; MATLAB/Simulink, Simscape, Multisim
Automation	Siemens TIA Portal (S7-1500, PLCSIM Advanced, WinCC), Rockwell Studio 5000, ladder logic, OpenPLC, SCADA, Modbus TCP; Process Simulate (virtual commissioning), UR (PolyScope), Cognex In-Sight
Testing & Software	HIL, LabVIEW, Python (OpenCV, YOLO), C#, Git/GitHub (basic)
Lab & CAD	Oscilloscope, function generator, DC power supply, multimeter; SolidWorks (CSWA), Siemens NX; turning, milling, MIG welding, 3D printing
Languages	Spanish (native), English (C1), German (B1), French (B1)

Certifications

- MathWorks: Simulink & Simscape Onramp (2026)
- Altium: Educational PCB Basic Design (2026)
- Dassault Systèmes: CSWA
- Siemens: Process Simulate Associate (2026)
- Universal Robots (4 tracks), MiR (3 courses) (2026)
- MIT.nano: Certificate of Participation (2025)